

Reasoning

Direction Distance

Level-1

Q1 Study the following information carefully and answer the questions given below:

Point R is 12m north of point Q. Point U is 5m west of V. Point V is 4m north of point S. Point X is 9m south of point V. Point S is 7m east of point R. Point X is 7m North of W.

How far is Point V with respect to Point W?

- (A) 12m (B) 9m
(C) 16m (D) 10m
(E) 20m

Q2 Study the following information carefully and answer the questions given below:

Point R is 12m north of point Q. Point U is 5m west of V. Point V is 4m north of point S. Point X is 9m south of point V. Point S is 7m east of point R. Point X is 7m North of W.

If Point A is the midpoint of RQ, what is the shortest distance and direction of point A with respect to Q?

- (A) 6m, North (B) 6m, South
(C) 16m, North (D) 12m, South
(E) None of these

Q3 Study the following information carefully and answer the questions given below:

Point R is 12m north of point Q. Point U is 5m west of V. Point V is 4m north of point S. Point X is 9m south of point V. Point S is 7m east of point R. Point X is 7m North of W.

What is the direction of point W with respect to X?

- (A) South (B) Southeast
(C) West (D) Northwest
(E) None of these

Q4 Study the following information carefully and answer the questions given below:

Point N is 10m east of point M. Point O is 15m north of point N. Point P is 15m east of O. Point Q is 15m north of point P. Point R is 15m west of point Q. Point S is 5m South of point R.

Find the distance between N and S?

- (A) 20m (B) 25m
(C) 15m (D) 35m
(E) 30m

Q5 Study the following information carefully and answer the questions given below:

Point N is 10m east of point M. Point O is 15m north of point N. Point P is 15m east of O. Point Q is 15m north of point P. Point R is 15m west of point Q. Point S is 5m South of point R.

If a person facing north at Point N moves 90 degree clockwise and move 10m further to reach Point Z then find the distance between Point Z and Point M?

- (A) 20m (B) 15m
(C) 25m (D) 35m
(E) 30m

Q6 Study the following information carefully and answer the questions given below:

Point N is 10m east of point M. Point O is 15m north of point N. Point P is 15m east of O. Point Q



is 15m north of point P. Point R is 15m west of point Q. Point S is 5m South of point R.

Q is in Which direction with respect to Point O?

- (A) Southeast (B) Northwest
(C) Northeast (D) West
(E) East

Q7 Study the following information carefully and answer the questions given below:

Point P is 20m to the south of Point Q. Point Q is 15m to the west of Point R. Point R is 40m to the north of Point S. Point S is exactly between Points A and B such that Points A, S and B form a straight line of 30m.

How far and in which direction is Point R with respect to Point S ?

- (A) South-20m (B) North-35m
(C) South-40m (D) North-40m
(E) None of these

Q8 Study the following information carefully and answer the questions given below:

Point P is 20m to the south of Point Q. Point Q is 15m to the west of Point R. Point R is 40m to the north of Point S. Point S is exactly between Points A and B such that Points A, S and B form a straight line of 30m.

What is the shortest distance between Point P and Point R?

- (A) 20m (B) 25m
(C) 15m (D) 30m
(E) None of these

Q9 Study the following information carefully and answer the questions given below:

Point T is 80 m towards the North of Point Q. Point R is 30 m towards the West of Point T. Point U is 40 m towards the South of Point R. Point S is

30 m towards the East of Point U. Point W is 30 towards the West of Point Q.

If a person walks 20 m towards East from Point U and takes a right turn and walk 40m, which of the following points he could reach first?

- (A) W (B) T
(C) Q (D) U
(E) None of these

Q10 Study the following information carefully and answer the questions given below:

Point T is 80 m towards the North of Point Q. Point R is 30 m towards the West of Point T. Point U is 40 m towards the South of Point R. Point S is 30 m towards the East of Point U. Point W is 30 towards the West of Point Q.

Q is in which direction with respect to S?

- (A) South (B) North
(C) East (D) West
(E) None of these

Q11 Study the following information carefully and answer the questions given below:

Point T is 80 m towards the North of Point Q. Point R is 30 m towards the West of Point T. Point U is 40 m towards the South of Point R. Point S is 30 m towards the East of Point U. Point W is 30 towards the West of Point Q.

Which of the following points can be joined by a straight line?

- (A) T, U, Q (B) U, W, R
(C) T, R, W (D) U, S, R
(E) None of these

Q12 Study the following information carefully and answer the questions given below:

Rohan is standing at Point O. He walks for 4m towards east and reaches Point P. He takes a left turn and walks for 2m to reach Point Q. He then



takes a right turn and walks 8m to reach Point R. He again takes a right turn and walks 10m to reach Point S.

From point S, Rohan walks 9m towards west to reach Point A. In which direction from Point A and for how many meter he needs to walk to reach 8 meter south of Point O.

- (A) South - 5m (B) West - 3m
(C) North - 4m (D) South - 3m
(E) None of these

Q13 Study the following information carefully and answer the questions given below:

Rohan is standing at Point O. He walks for 4m towards east and reaches Point P. He takes a left turn and walks for 2m to reach Point Q. He then takes a right turn and walks 8m to reach Point R. He again takes a right turn and walks 10m to reach Point S.

If Point Z is 4m to the west of Point Q, what is the distance between Point O and Point Z ?

- (A) 2m (B) 3m
(C) 4m (D) 5m
(E) None of these

Q14 Study the following information carefully and answer the questions given below:

Point A is 18m to the north of Point B. Point D is west of point F. Point C is 20m to the west of Point A. Point D is 15m to the south of point C. Point E is exactly midway between Points D and F in such a manner that Point D, E and F form a horizontal straight line of 40m. Point G is 15m to the north of Point F.

How far and in which direction is Point G from Point A ?

- (A) 18m towards North
(B) 20m towards East
(C) 15m towards North

- (D) 40m towards West
(E) None of these

Q15 Study the following information carefully and answer the questions given below:

Point A is 18m to the north of Point B. Point D is west of point F. Point C is 20m to the west of Point A. Point D is 15m to the south of point C. Point E is exactly midway between Points D and F in such a manner that Point D, E and F form a horizontal straight line of 40m. Point G is 15m to the north of Point F.

In which of the following direction is Point B with respect to Point C ?

- (A) South-East (B) South
(C) East (D) South-West
(E) None of these

Q16 Study the following information carefully and answer the questions given below:

Point A is 18m to the north of Point B. Point D is west of point F. Point C is 20m to the west of Point A. Point D is 15m to the south of point C. Point E is exactly midway between Points D and F in such a manner that Point D, E and F form a horizontal straight line of 40m. Point G is 15m to the north of Point F.

What is the shortest distance between Point A and Point D?

- (A) 20m (B) 30m
(C) 25m (D) 15m
(E) None of these

Q17 Study the following information carefully and answer the questions given below:

Point A is 18m to the South of Point B. Point C is 20m to the East of Point A. Point D is 16m to the South of point C. Point E is exactly midway between Points D and C in such a manner that



Point D, E and C form a vertical line. Point G is 18m to the north of Point C.

Which of the following Points form a straight line?

- (A) B, G, D (B) A, C, G
(C) G, C, D (D) B, C, D
(E) None of these

Q18 Study the following information carefully and answer the questions given below:

Point A is 18m to the South of Point B. Point C is 20m to the East of Point A. Point D is 16m to the South of point C. Point E is exactly midway between Points D and C in such a manner that Point D, E and C form a vertical line. Point G is 18m to the north of Point C.

In which direction is Point G from Point A ?

- (A) North (B) South-East
(C) North-East (D) South
(E) None of these

Q19 Study the following information carefully and answer the questions given below:

Point A is 18m to the South of Point B. Point C is 20m to the East of Point A. Point D is 16m to the South of point C. Point E is exactly midway between Points D and C in such a manner that Point D, E and C form a vertical line. Point G is 18m to the north of Point C.

What is the distance between Point G and Point D?

- (A) 30m (B) 34m
(C) 20m (D) 18m
(E) None of these

Q20 Study the following information carefully and answer the questions given below:

Point A is 18 m to the south of Point B. Point B is 10 m to the east of Point C. Point C is 8 m to the north of Point D. Point E is 6 m west of Point D.

Point F is 14 m south of Point E. Point G is 16 m east of Point F.

What is the direction of G with respect of E?

- (A) North-East
(B) South-East
(C) South-West
(D) Cannot be determined
(E) None of these

Q21 Study the following information carefully and answer the questions given below:

Point A is 18 m to the south of Point B. Point B is 10 m to the east of Point C. Point C is 8 m to the north of Point D. Point E is 6 m west of Point D. Point F is 14 m south of Point E. Point G is 16 m east of Point F.

What is the shortest distance between Point E and Point C?

- (A) 20m (B) 10m
(C) 25m (D) 15m
(E) None of these

Q22 Study the following information carefully and answer the questions given below:

Point S1 is north of Point S2. Point S3 is north of Point S4 and west of Point S5. Point S6 is east of Point S1 and south of Point S5. Point S7 is west of Point S2. Point S1 is south-west of Point S4 which is north of Point S8. Point S8 is west of Point S6 but not exactly middle of Point S1 and S6.

If the distance between S7 and S2 is 12km and the distance between S1 and S2 is 5km then what is the shortest distance between S7 and S1?

- (A) 17km (B) 16km
(C) 13km (D) 8km
(E) None of these

Q23 Study the following information carefully and answer the questions given below:



Point S1 is north of Point S2. Point S3 is north of Point S4 and west of Point S5. Point S6 is east of Point S1 and south of Point S5. Point S7 is west of Point S2. Point S1 is south-west of Point S4 which is north of Point S8. Point S8 is west of Point S6 but not exactly middle of Point S1 and S6.

Which of the following point is north-east of Point S4?

- (A) Point S6 (B) Point S7
(C) Point S5 (D) Point S1
(E) None of these

Q24 Study the following information carefully and answer the questions given below:

Sunil starts walking towards East from Point A. He walks for 20m and then takes a left turn. He then walks for 10m and again takes a left turn. He reaches Point B after walking for 5m. He takes a right turn, walk for 6m and reaches Point C. He finally stops at Point D after walking for 8m to the left of Point C.

How far is Point D from Point B ?

- (A) 11m (B) 7m
(C) 10m (D) 18m
(E) None of these

Q25 Study the following information carefully and answer the questions given below:

Sunil starts walking towards East from Point A. He walks for 20m and then takes a left turn. He then walks for 10m and again takes a left turn. He reaches Point B after walking for 5m. He takes a right turn, walk for 6m and reaches Point C. He finally stops at Point D after walking for 8m to the left of Point C.

If, a person walks 7m North from Point A, takes a right turn and stops after walking for 13m. Then, in which direction is Point D with respect to his final position ?

- (A) North-West (B) North-East
(C) South (D) East
(E) None of these

Q26 Study the following information carefully and answer the questions given below:

Raman drove his car for 50 km to North from his house. Then he turned right and drove 80km. Again he turned left and drove for yet another 60km to reach Hospital. From Hospital he drove his car for 60km to East. Then he turned left and drove 50km. Again he turned left and drove 60km to reach his Office.

What is the shortest distance between his Office from the Hospital?

- (A) 60km (B) 50km
(C) 70km (D) 80km
(E) None of these

Q27 Study the following information carefully and answer the questions given below:

Raman drove his car for 50km due North from his house. Then he turned right and drove 80km. Again he turned left and drove for yet another 60km to reach Hospital. From Hospital he drove his car for 60km due East. Then he turned left and drove 50km. Again he turned left and drove 60km to reach his Office.

With respect to the hospital, What is the direction of the Raman's house?

- (A) North-East (B) North-West
(C) South (D) South-West
(E) None of these

Q28 Study the following information carefully and answer the questions given below:

Point P is 30 m to the east of Point Q. Point R is 10 m to the south of Point P. Point S is 15 m to the west of Point R. Point T is exactly in the middle of the Points S and U. Points S, T and U lie in a



straight line. The length of the line STU is 20 meters. Point U is to the north Point S. Point V is 15 m to the east of Point U.

How far and in which direction is Point V from Point P ?

- (A) 10 m, North (B) 15 m, South
(C) 10 m, East (D) 15 m, West
(E) None of these

Q29 Study the following information carefully and answer the questions given below:

Point P is 30 m to the east of Point Q. Point R is 10 m to the south of Point P. Point S is 15 m to the west of Point R. Point T is exactly in the middle of the Points S and U. Points S, T and U lie in a straight line. The length of the line STU is 20 meters. Point U is to the north Point S. Point V is 15 m to the east of Point U.

How far is Point Q from Point P ?

- (A) 20m (B) 25m
(C) 15m (D) 30m
(E) None of these

Q30 Study the following information carefully and answer the questions given below:

Point P is 30 m to the east of Point Q. Point R is 10 m to the south of Point P. Point S is 15 m to the west of Point R. Point T is exactly in the middle of the Points S and U. Points S, T and U lie in a straight line. The length of the line STU is 20 meters. Point U is to the north Point S. Point V is 15 m to the east of Point U.

If U is the middle Point of T and W than in which direction is Point S from Point W ?

- (A) North (B) East
(C) South (D) West
(E) None of these



Answer Key

Q1 (C)
Q2 (A)
Q3 (A)
Q4 (B)
Q5 (A)
Q6 (C)
Q7 (D)
Q8 (B)
Q9 (C)
Q10 (A)
Q11 (B)
Q12 (B)
Q13 (A)
Q14 (B)
Q15 (A)

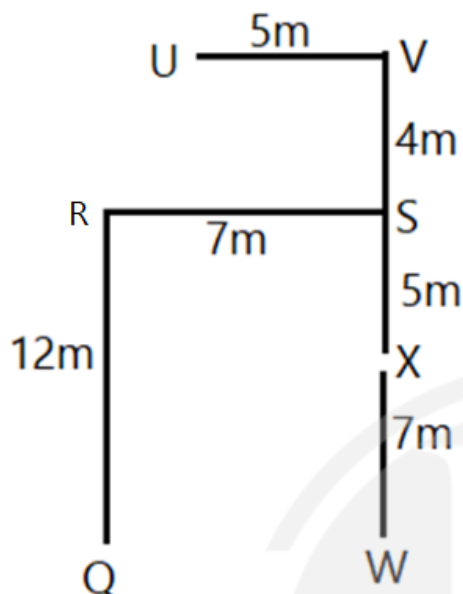
Q16 (C)
Q17 (C)
Q18 (C)
Q19 (B)
Q20 (B)
Q21 (B)
Q22 (C)
Q23 (C)
Q24 (C)
Q25 (A)
Q26 (B)
Q27 (D)
Q28 (A)
Q29 (D)
Q30 (C)



Hints & Solutions

Q1 Text Solution:

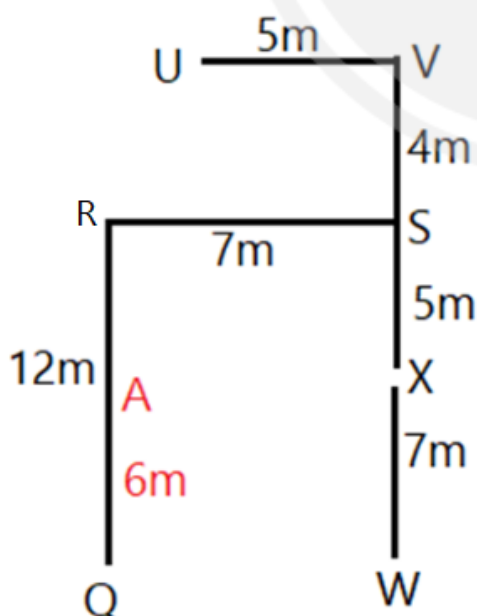
According to the question, we can draw the diagram-



16m distance of Point V with respect to Point W.

Q2 Text Solution:

According to the question, we can draw the diagram-

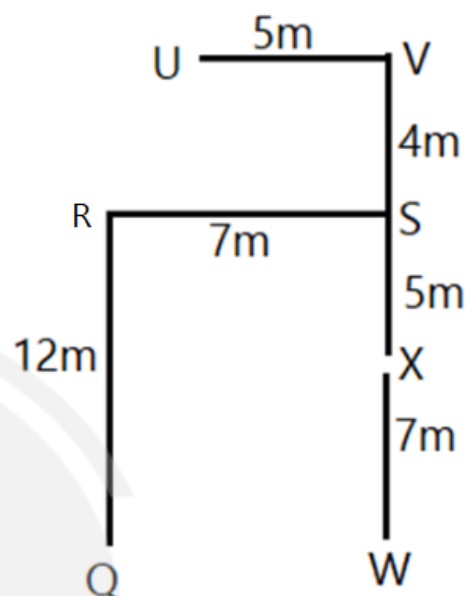


If Point A is the midpoint of RQ, so the shortest distance and direction of point A with respect to

Q is 6m.

Q3 Text Solution:

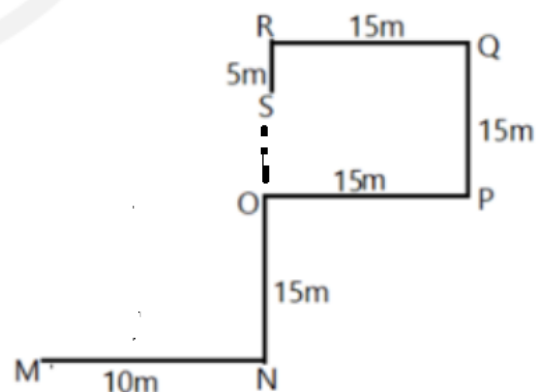
According to the question, we can draw the diagram-



The direction of point W with respect to X is South.

Q4 Text Solution:

According to the question, we can draw the diagram-

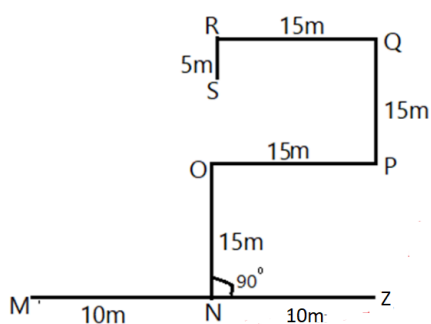


The distance between N and S = 25m

Q5 Text Solution:

According to the question, we can draw the diagram-

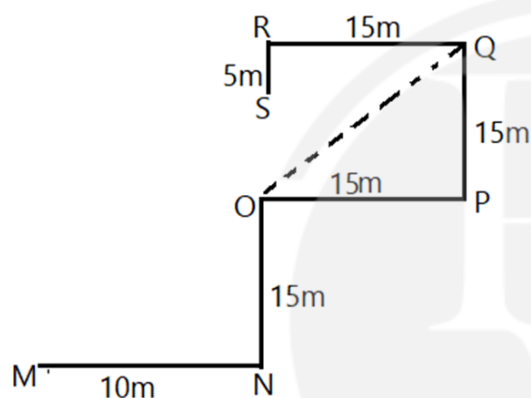




Distance between Point Z and Point M is 20m.

Q6 Text Solution:

According to the question, we can draw the diagram-

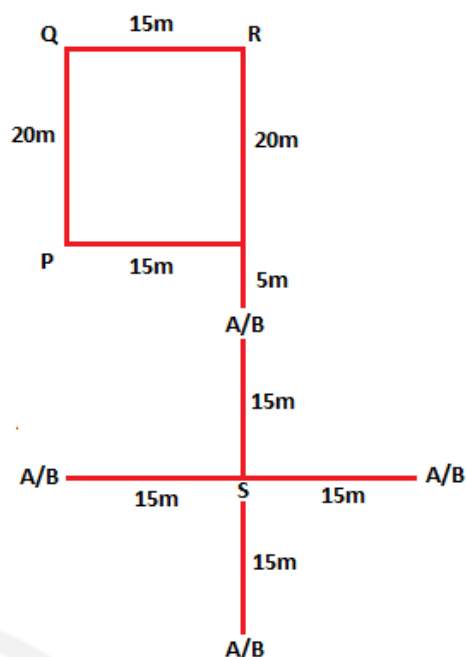


Q is in the Northeast direction with respect to Point O.

Hence, option c is the correct answer.

Q7 Text Solution:

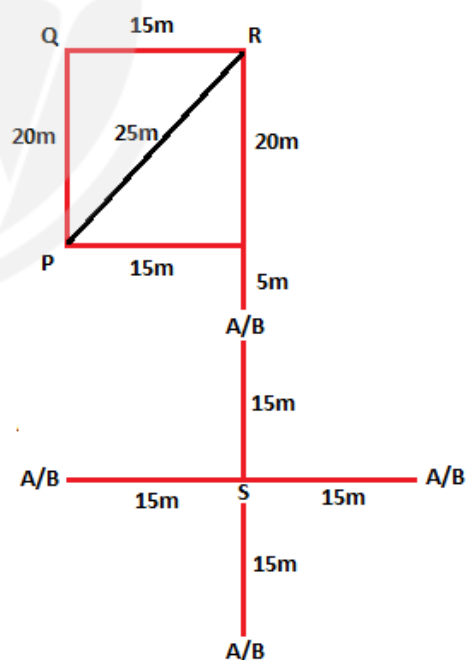
According to the question, we can draw the diagram-



We can clearly see that the Point R is in the North of Point S and the distance between RS is 40m.

Q8 Text Solution:

According to the question, we can draw the diagram-



$$PR^2 = 15^2 + 20^2 = 225 + 400 = 625$$

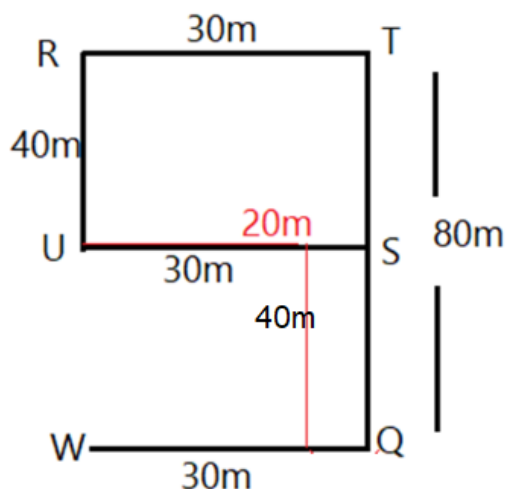
So, $PR = 25$

We can clearly see, the distance between Point P and Point R = 25m



Q9 Text Solution:

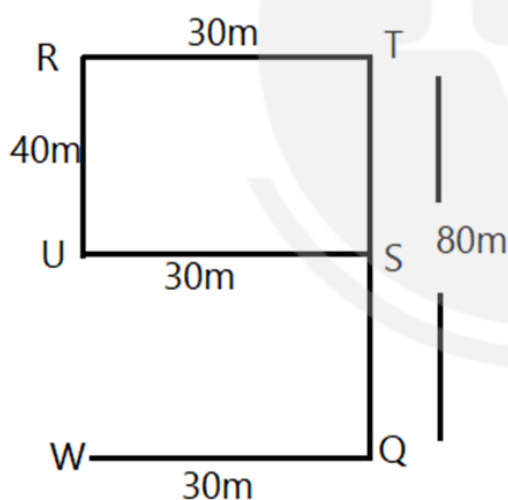
According to the question, we can draw the diagram-



Q Point he could reach first.

Q10 Text Solution:

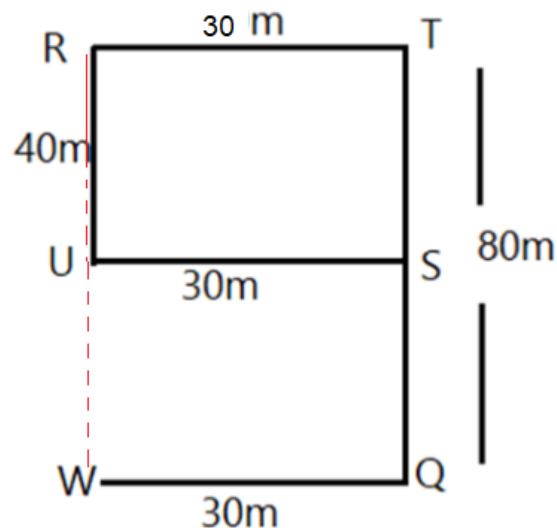
According to the question, we can draw the diagram-



Q is in the South direction with respect to S.

Q11 Text Solution:

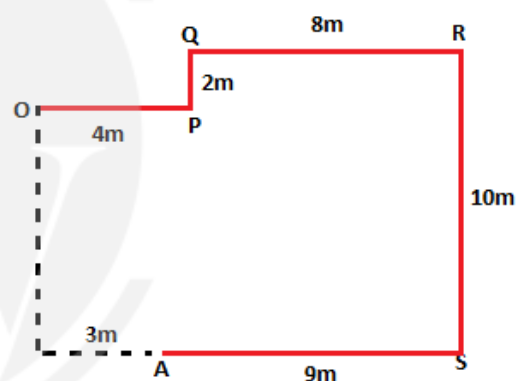
According to the question, we can draw the diagram-



U, W, R points are in a straight line.

Q12 Text Solution:

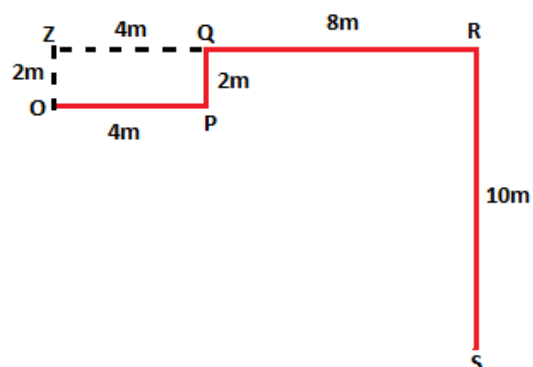
According to the question, we can draw the diagram-



Now, Rohan has to walk 3 m to the west from Point A to reach 8 m south of Point O.

Q13 Text Solution:

According to the question, we can draw the diagram-

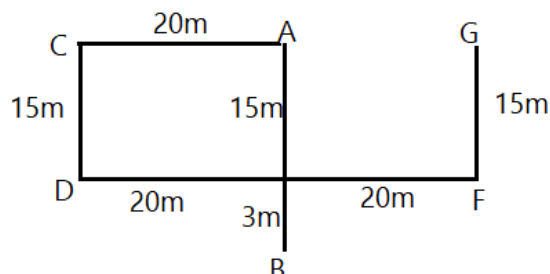


We can clearly see the distance between O and Z is 2m.

OZ = 2 metre

Q14 Text Solution:

According to the question, we can draw the diagram-

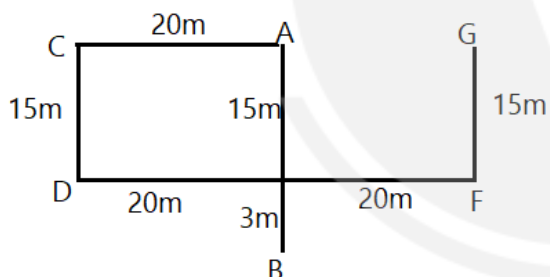


So, we can clearly see that 20m and in east direction is Point G from Point A.

Hence, option b is the correct answer.

Q15 Text Solution:

According to the question, we can draw the diagram-

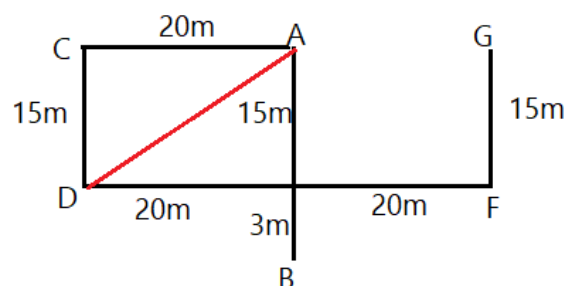


Point B is in South-East direction with respect to Point C.

Hence, option a is the correct answer.

Q16 Text Solution:

According to the question, we can draw the diagram-



The shortest distance between Point A and Point D is

$$\sqrt{20^2 + 15^2}$$

$$\sqrt{400 + 225}$$

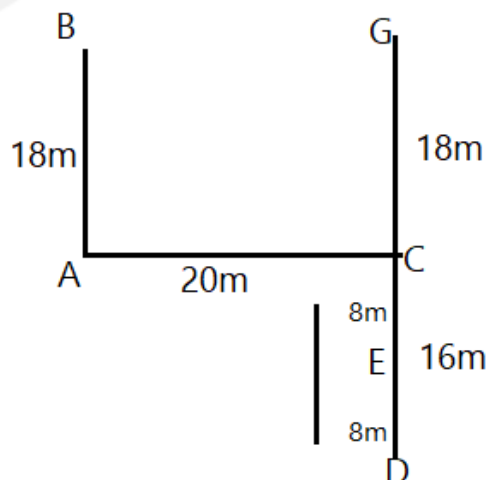
$$\sqrt{625}$$

= 25 m

Hence, option c is the correct answer.

Q17 Text Solution:

According to the question, we can draw the diagram-



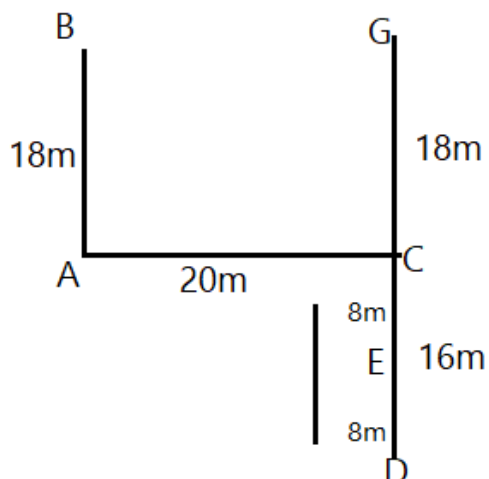
Points G, C, D are in the same line.

Hence, option c is the correct answer.



Q18 Text Solution:

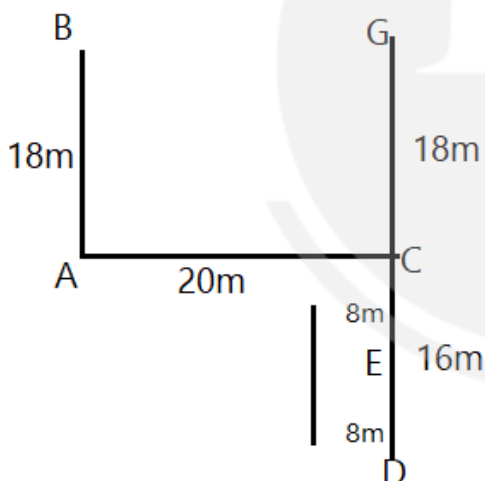
According to the question, we can draw the diagram-



So, we can clearly see that in the North-East direction is Point G from Point A.

Q19 Text Solution:

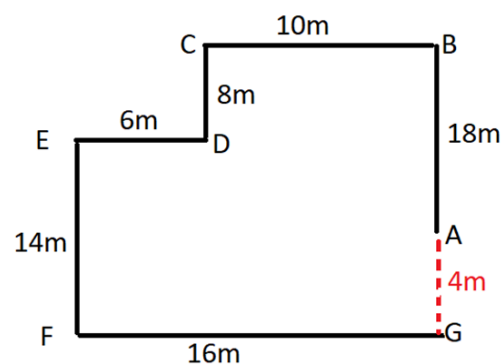
According to the question, we can draw the diagram-



So, we can clearly see that in the distance between Point G and Point D is 34m

Q20 Text Solution:

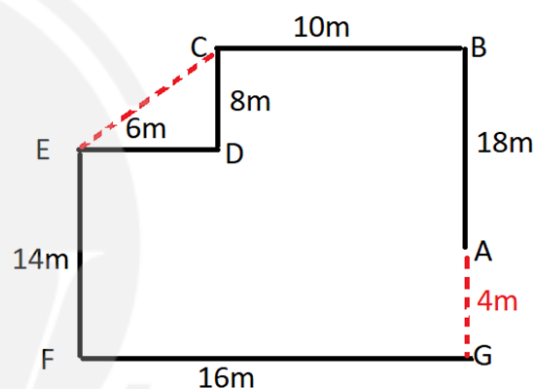
According to the question, we can draw the diagram-



We can clearly see that the direction of Point G with respect of E is South-East.

Q21 Text Solution:

According to the question, we can draw the diagram-



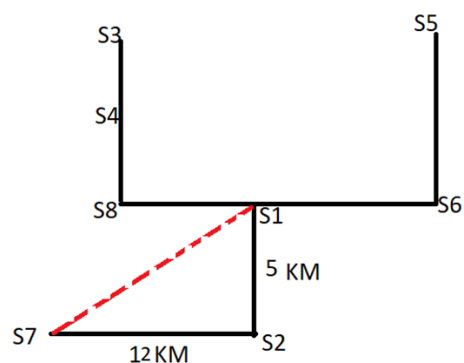
The distance between Point E and Point C is

$$\begin{aligned} EC &= \sqrt{(8^2 + 6^2)} \\ &= \sqrt{64 + 36} \\ &= \sqrt{100} \\ &= 10 \text{ m} \end{aligned}$$

Q22 Text Solution:

According to the question, we can draw the diagram-



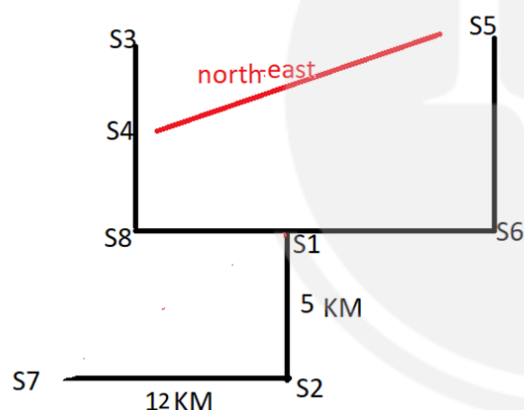


The distance between S7 and S1 =

$$\begin{aligned} & \sqrt{12^2 + 5^2} \\ & \sqrt{144 + 25} \\ & \sqrt{169} \\ & = 13\text{km} \end{aligned}$$

Q23 Text Solution:

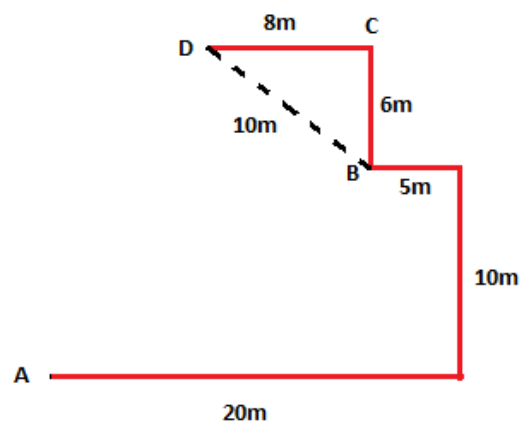
According to the question, we can draw the diagram-



We can clearly see that only S5 is North-East of S4.

Q24 Text Solution:

According to the question, we can draw the diagram-



Using Pythagoras Theorem:

$$BD^2 = CD^2 + BC^2$$

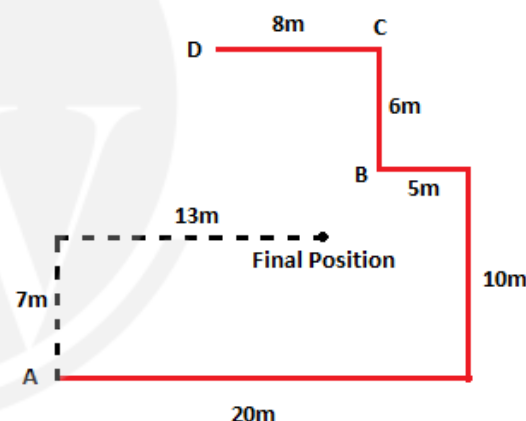
$$BD^2 = 8^2 + 6^2 = 100$$

Therefore, $BD = 10\text{m}$

We can clearly see, the distance between D and B is 10m.

Q25 Text Solution:

According to the question, we can draw the diagram-

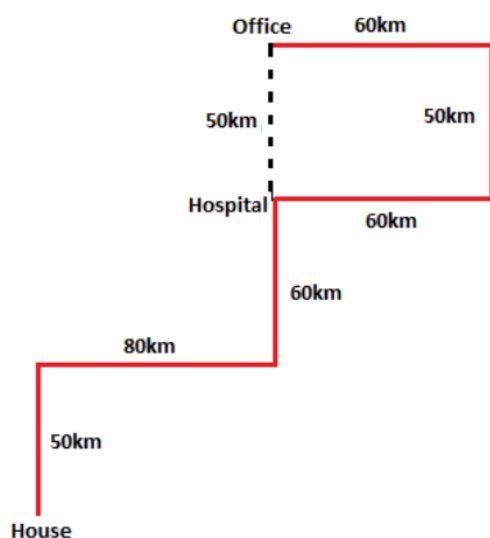


We can clearly see, the position of D is in North-West direction.

Q26 Text Solution:

According to the question, we can draw the diagram-

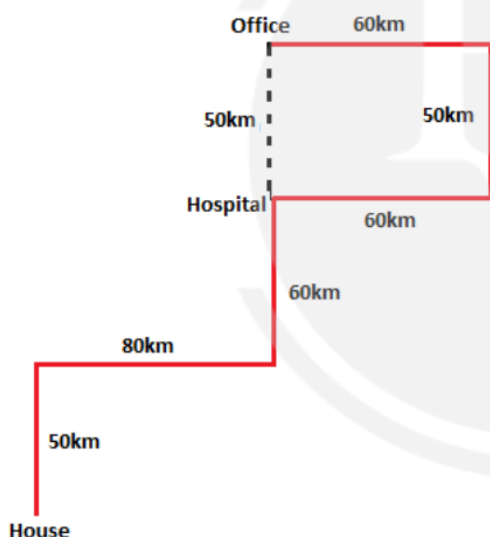




We can clearly see, the distance between Hospital and Office is 50km.

Q27 Text Solution:

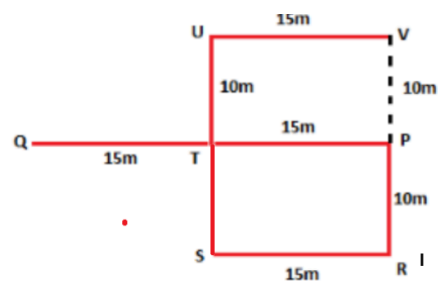
According to the question, we can draw the diagram-



With respect to the hospital, the direction of the Raman's house is in South-West.

Q28 Text Solution:

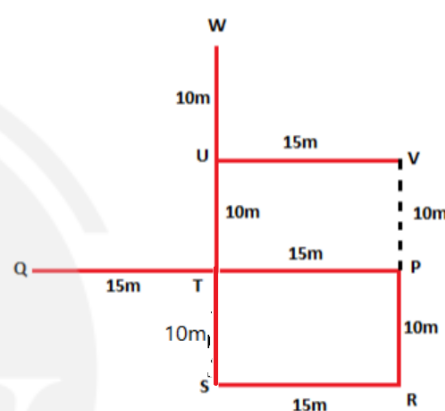
According to the question, we can draw the diagram-



We can clearly see, Point V is 10m North of Point P.

Q29 Text Solution:

According to the question, we can draw the diagram-



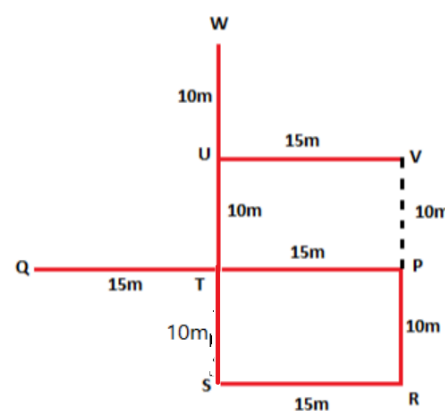
We can clearly see, the distance between Point Q and Point P = QT + TP

$$QP = 15 + 15 = 30m$$

Hence, option(d) is the correct answer.

Q30 Text Solution:

According to the question, we can draw the diagram-



We can clearly see, Point S is South to the Point W.



Hence, option(c) is the correct answer.



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